Aalto University, School of Science
Master's Programme in Computer, Communication and Information Sciences

Master's thesis plan

Scope Attribution for Static UI Transition Graph Extraction
in Single-Page Applications

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Major	Computer Science
Extent	30 ECTS
Language	English
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1 Topic

A UI transition graph models an application as a set of screens and the transitions a user can trigger between them. Static extraction of such graphs from source code is well established for Android but has not been studied for client-side web frameworks, which are modelled dynamically by crawling. The thesis measures static extraction on real Angular applications using an existing open-source analyser developed by the student, classifies where it fails, and proposes and evaluates an analysis, called scope attribution, that addresses the largest class of failures: navigation sites located in shared components, injected services and route guards, which the current extractor attributes to no screen.

2 Research questions

RQ1 How much of the navigation in real Angular applications can be extracted statically? Measured as the fraction of navigation constructs in the source that become deterministic transitions in the graph.

RQ2 Where does extraction fail, and why? Answered by a defect classification in which each class must be confirmed by at least two independent evaluation methods.

RQ3 Does scope attribution recover the largest defect class? Measured as the change in the RQ1 ratio between two fixed tool versions on identical inputs, with precision on the deterministic tier not decreasing.

RQ4 Does the resulting graph detect source changes well enough to support change impact analysis? Measured by mutation testing.

The thesis does not claim completeness, which is undecidable for dynamic navigation; it proves conditional soundness over a declared scope and measures recall. It makes no claim about the effect of the graph on coding agents, and no claim beyond Angular.

3 Material and methods

Corpus. Six open-source Angular applications, selected by six published criteria from ten candidates, pinned to exact commits, and spanning 14 to 674 declared routes. Threats to the validity of an open-source corpus and their mitigations are addressed in the thesis. An industrial case study on a proprietary Angular application, reporting aggregate figures only, is proposed as an extension subject to the supervisor's view and the employer's permission.

Evaluation. Four independent methods, each with a declared blind spot: verification of every transition against its source witness (precision); comparison with an independently written route parser (diagnosis); the applications' own passing end-to-end tests as ground truth (recall); and mutation testing (sensitivity). The whole evaluation is reproducible from one command with full provenance.

Preliminary results. On the current extractor, static extraction recovers between 0 and 14 percent of navigation constructs as deterministic transitions, and three of six applications yield routes but no transitions. Where deterministic transitions are produced they are almost always correct (75 of 76 verified against source). Three independent methods point to attribution as the dominant cause of loss.

4 Planned structure

	Chapter	Pages
1	Introduction: motivation, research questions and scope, thesis structure	4–6
2	Background: transition graphs; static extraction for Android; navigation models for web applications; dynamic models of single-page applications; Angular routing and dependency injection; formal foundations	15–20
3	Research material and methods: corpus and validity; graph model; scope attribution; implementation; evaluation protocol	15–20
4	Results, one section per research question	12–15
5	Discussion: findings, limitations, declared approximations, future work	5–8
6	Conclusion	1–2

Target length 60 to 75 pages. Chapters 1 to 3 are drafted.

5 Schedule

Date	Milestone	Completion criterion
28 Sep	Baseline frozen	Tool version tagged; corpus re-measured; result tables generated.
5 Oct	Chapter 3 final	Corpus, model and protocol sections complete.
12 Oct	Attribution implemented	Component nesting, router outlets, DI-aware call graph, guards, anchored edges.
15 Oct	Checkpoint	If attribution is incomplete, RQ4 moves to future work and RQ3 reports the implemented subset.
19 Oct	Second measurement	Same corpus and harness on the new tag; before and after tables.
26 Oct	Chapters 2, 4, 5 drafted	
2 Nov	Complete draft to supervisor	
November	Thesis presentation; revisions	Date agreed with supervisor.
30 Nov	Submission	PDF/A and abstract in MyStudies; graduation request in Sisu.

The student's study right ends on 31 December 2026. The schedule therefore targets the December meeting of the Degree Programme Committee. The study coordinator will be consulted on the exact date the study right must cover and on the extension procedure.

6 Risks

Risk	Response
Attribution not complete by 12 October	The empirical study (RQ1, RQ2) stands alone. RQ4 is dropped first; RQ3 reports what is implemented and declares the rest.
Supervisor reading time in November	Draft date and reading window agreed in advance.
Corpus applications require a backend to run	No planned method executes an application.
Tool changes during measurement	Measurements are taken only against tagged versions.
Study right timing	Study coordinator consulted; extension application filed if advised.

7 Deliverables

The thesis; the analyser with two tagged versions corresponding to before and after attribution; the corpus manifest and the evaluation harness; the generated results with provenance; a verified bibliography.

Approval

Duy Khanh Tran, student	Prof. Petri Vuorimaa, supervisor
Date: _____	Date: _____